

Muhammad Saad Ibrahim

Software Engineer

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Profile

Experienced Software Engineer specializing in event-driven architectures, event sourcing, and CQRS patterns. Proficient in full-stack development with expertise in TypeScript, F#, and modern frontend frameworks. Passionate about Clean Architecture, Domain-Driven Design, and building scalable distributed systems.

Education

Kafrelsheikh University

Bachelor's Degree in Computer Science – Information Systems Department, CGPA: 2.95

Egypt

Aug 2020 – June 2024

Experience

Software Engineer

Jul 2025 – Present

@Bytes Maestros

Stockholm County, Sweden (Remote)

- Developed an e-commerce platform from scratch using TypeScript and SvelteKit with Clean Architecture and Domain-Driven Design, implementing event-driven microservices.
- Built event-sourced systems using NATS JetStream and PostgreSQL event stores with CQRS pattern, implementing event consumers for order management, product catalog, and marketing features.
- Developed SvelteKit frontends with TailwindCSS, Storybook component library, and comprehensive testing (Vitest, Playwright), including internationalization support.
- Built a B2B SaaS restaurant management system using F# and .NET 10 with Oxpecker web framework, applying Functional Core Imperative Shell pattern and schema-first API design with JSON Schema TypeProviders.
- Designed and implemented PostgreSQL databases with Kysely query builder, type-safe migrations, optimistic concurrency control, and multi-tenancy support.
- Implemented observability-first architecture with causation/correlation IDs for distributed tracing, type-safe validation using Zod and functional programming patterns.

Backend Developer

Nov 2024 – Jun 2025

@Techno Island, Software House

Maadi, Cairo

- Developed 3 modules from scratch for the *E-Clinic* SaaS application.
- Developed 6 modules from scratch for the *NewLoanz* third-party banking CRM system.
- Worked on the Transactions Allocation background job and managed the Promises to Pay workflow.
- Integrated the *NewLoanz* API with internal and external jobs to process CSV files from the Core Banking System (CBS).
- Integrated an online API to retrieve daily currency exchange rates.
- Analyzed user stories, BRD documents, and Figma designs to gather requirements.

Backend Developer

Feb 2023 – Aug 2024

@Zellige, Product-Based Startup

Alexandria, Egypt

- Designed and developed over 45 features, delivering them to customers.
- Refactored and optimized the existing codebase, resolving bugs efficiently.
- Conducted end-to-end testing with front-end developers to ensure a seamless user experience.

Projects

Custom Shell | Codecrafters Challenge 🏆

May 2025

- Developed a custom shell in C# as part of the Codecrafters challenge.
- Implemented core built-in commands (echo, pwd, exit, type) and support for external commands via environment variables.
- Enabled directory navigation with cd and implemented robust quoting (single, double, and escape characters).
- Added output/error redirection (including append mode) and advanced command auto-completion.
- Built modular components: custom input reader (Tab, Backspace, Enter), state machine tokenizer, token parser, and a unified command executor.

- Led a team of nine developers, ensuring smooth collaboration and efficient workflows.
- Developed a robust backend with ASP.NET Core, creating over 70 APIs for mobile and web clients.
- Designed a comprehensive ERD with 20+ tables, following best practices in database design.
- Analyzed client needs to ensure the project met all requirements.
- *Technologies: ASP.NET 8 Web APIs, EF Core, Microsoft SQL Server, Docker, Swagger.*

Technical Skills

Languages: C#, F#, TypeScript, SQL

Frameworks: ASP.NET Core, .NET 10, Entity Framework Core, Dapper, LINQ, Oxpecker, SvelteKit

Frontend: Svelte, TailwindCSS, Flowbite, Vite, Google Maps API, Storybook

Databases: Microsoft SQL Server, PostgreSQL, MongoDB

Event Streaming & Messaging: NATS JetStream, RabbitMQ, MassTransit

ORM/Query Builders: Kysely, Entity Framework Core, Dapper, Npgsql.FSharp

Methodologies: Clean Architecture, Domain-Driven Design, Event Sourcing, CQRS, Functional Core Imperative Shell, Modular Monolith, OOP, SOLID Principles, RESTful API Design, Schema-First API Design, ERD Design, Event-Driven Architecture, Multi-tenancy

Developer Tools: Git, GitHub, Docker, Docker Compose, Azure Boards, Rider, pnpm, Moon Build System

Testing: Expecto, Faqt, XUnit

Libraries & Tools: Hangfire, FluentResults, Zod, Serilog, Sentry, Better Auth, Inlang/Paraglide

Certificates

- ACM ECPC Contest: Qualified for the ECPC Finals in the 2023 season after securing second place in the university qualification round.